Abdullah Hendy

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Education

University of Michigan - Ann Arbor | *GPA: 3.75* — *Aug 2024 - May 2026*
Master's of Science | *Electrical and Computer Engineering* — *Ann Arbor, MI*

California State University - Northridge | *GPA: 4.0* — *Aug 2019 - May 2024*
Bachelor's of Science | *Computer Engineering* — *Northridge, CA*
Minor | *Mathematics*

Experience

R&D Engineer — *Oct 2023 - Aug 2025*
Keysight Technologies — *Calabasas, CA*

- Designed and developed custom FPGA-based hardware leveraging PCIe 5.0 for high-bandwidth data movement, achieving 2x throughput and 2x reduction in latency for AI/HPC infrastructure.
- Modernized legacy synchronization infrastructure by architecting multiple gRPC microservices and custom low-level FPGA logic, achieving nanosecond-level timing precision for synchronizing distributed Ethernet test devices.
- Developed a P4 based packet capture module within a programmable switch pipeline, enabling low-level traffic visibility for debugging high-speed Ethernet designs.

Applications Engineer - Intern — *Jun 2023 - Aug 2023*
Cirrus Logic — *Austin, TX*

- Developed a Python API wrapper for multi-FPGA silicon emulation to support internal validation and external customer verification, reducing workflow setup time and standardizing protocols across diverse platforms.
- Characterized dynamic supply rail tracking on Class H amplifiers to address customer inquiries, providing detailed signal analysis to verify expected power draw under various load conditions.

Projects

Mars Mission Emulation (Computer Vision and Robotics) | ***Finalist, OpenCV AI 2023 Competition***

- Engineered a drone-to-robot system emulating NASA Mars mission constraints; The drone integrates a Raspberry Pi and Coral TPU for MobileNet-SSD object detection, transmitting navigation data to the robot via a custom Bluetooth commands using Serial Port Profile (SPP).
- Developed bare-metal embedded C firmware for the MSP432-based ground robot, including a FSM to orchestrate mission tasks; implemented register-level drivers, digital signal filtering, and a PID controller for autonomous navigation.

Real-Time Speech Translation Engine (NLP and Machine Learning)

- Built a low-latency multilingual translation engine integrating Faster Whisper and Opus-MT models; achieved real-time streaming through WebSockets and Opus codec compression.
- Designed the server-side inference to support concurrent AR/mobile clients with minimal overhead.

Class AB Audio Amplifier Speaker (Analog Circuit Design)

- Designed, assembled, and tested a 5W Class AB amplifier PCB in KiCad, performing rigorous thermal modeling and tolerance analysis to ensure reliability; integrated TVS protection against inductive transients and optimized power operating points to minimize Total Harmonic Distortion.

Skills

Expertise: Machine Learning, Computer Vision, Natural Language Processing, Retrieval-Augmented Generation, FPGA, Embedded Systems, Microservices, Virtualization, Ethernet, CI/CD
Software: PyTorch, CUDA, OpenCV, LangChain, PSpice, Vivado, Vitis, KiCad, Unity, ROS, Docker, Kubernetes, Helm, gRPC, JIRA, Git, Artifactory, Wireshark, Flutter, Linux/Unix
Languages: Python, C/C++, Go, Java, Dart, Julia, MATLAB, ARM Assembly, VHDL, Verilog, P4